



JJ COLLEGE
ENGINEERING AND TECHNOLOGY
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J.J. College of Engineering and Technology

Department of Mechanical Engineering

ME25CO8

Metallurgy and Materials Science

Student Notes

Unit - III: Ferrous and Non-Ferrous Metals

Scheduled Hours: 9 Periods

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Course Handling Faculty

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Course Objectives

1. To learn the construction of phase diagrams and the use of the Iron–Iron carbide phase diagram for understanding microstructure formation.
2. To learn the selection and application of various heat treatment processes and their influence on microstructure formation.
3. To illustrate the different types of ferrous and non-ferrous alloys and their uses in the engineering field.
4. To illustrate different polymers, ceramics and composites and their uses in the engineering field.
5. To learn various testing procedures and failure mechanisms in engineering materials.

Full Course Content

Course Code: ME25CO8

Course Title: Metallurgy and Materials Science

Unit	Content
UNIT I	CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS Constitution of alloys – solid solutions, substitutional and interstitial phase diagrams, isomorphous, eutectic, eutectoid, peritectic and peritectoid reactions, Iron–Iron carbide equilibrium diagram. Classification of steel and cast-iron microstructure, properties and application.
UNIT II	HEAT TREATMENT Definition – full annealing, stress relief, recrystallisation and spheroidising – normalizing, hardening and tempering of steel. Isothermal transformation diagrams – cooling curves superimposed on I.T. diagram – continuous cooling transformation (CCT) diagram – austempering, martempering – hardenability, Jominy end quench test – case hardening, carburizing, nitriding, cyaniding, carbonitriding – flame and induction hardening – vacuum and plasma hardening – thermo-mechanical treatments – elementary ideas on sintering.
UNIT III	FERROUS AND NON-FERROUS METALS Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V, Ti and W) – stainless and tool steels – HSLA – maraging steels – grey, white, malleable, spheroidal alloy cast irons, copper and its alloys – brass, bronze and cupronickel – aluminium and its alloys; Al–Cu precipitation strengthening treatment – titanium alloys, Mg-alloys, Ni-based super alloys – shape memory alloys – properties and applications – overview of materials standards.
UNIT IV	NON-METALLIC MATERIALS Polymers – types of polymers, commodity and engineering polymers – properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK, PTFE, thermoset polymers, urea and phenol formaldehydes – nylon, engineering ceramics – properties and applications of Al ₂ O ₃ , SiC, Si ₃ N ₄ , PSZ and SIALON – intermetallics – composites – matrix and reinforcement materials – applications of composites – nano composites.
UNIT V	MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS Mechanisms of plastic deformation, slip and twinning – types of fracture – fracture mechanics – Griffith's theory – testing of materials under tension, compression and shear loads – hardness tests (Brinell, Vickers and Rockwell), micro and nano-hardness tests, impact test Izod and Charpy, fatigue and creep failure mechanisms.

Total: 45 periods (9 periods each)

Course Outcomes

At the end of the course, the students would be able to:

1. Explain alloys and phase diagrams, the Iron–Iron carbon diagram and steel classification.
2. Explain isothermal transformation, continuous cooling diagrams and different heat treatment processes.
3. Clarify the effect of alloying elements on ferrous and non-ferrous metals.
4. Summarize the properties and applications of non-metallic materials.
5. Explain the testing of mechanical properties.

Unit III Learning Focus

For examination preparation, Unit III should be studied with the following sequence:

1. **Alloying of steel:** role of Mn, Si, Cr, Mo, Ni, V, Ti and W in changing hardenability, corrosion resistance, toughness, wear resistance and high-temperature behaviour.
2. **Special steels:** stainless steels, tool steels, HSLA steels and maraging steels.
3. **Cast irons:** influence of graphite/carbide form on grey, white, malleable, spheroidal and alloy cast irons.
4. **Copper alloys:** brass, bronze and cupronickel, with properties and applications.
5. **Aluminium alloys:** wrought alloy series, important alloying elements and Al-Cu precipitation strengthening.
6. **Lightweight and high-performance alloys:** titanium alloys, magnesium alloys and nickel-based superalloys.
7. **Functional materials:** shape memory effect and superelasticity in shape memory alloys.
8. **Materials standards:** purpose of grades, specifications, test methods and traceability in engineering practice.

Engineering Context, Indian Relevance and Innovation Link

Why this unit matters: Material selection is rarely based on strength alone. A mechanical engineer must match composition, microstructure, processing, properties, service environment, cost and standards to the component requirement.

Indian manufacturing relevance

- Alloy steels, stainless steels and cast irons are widely used in automotive, railway, agricultural, pump, valve, machine-tool and infrastructure components.
- Aluminium and magnesium alloys support lightweight transport, electric mobility, consumer products and energy-efficient design.
- Titanium and nickel-based alloys are important in aerospace, chemical, marine, power-generation and high-temperature systems.
- Copper alloys remain important for electrical, thermal, marine and bearing applications.
- Materials standards, heat/batch traceability and test certificates connect classroom metallurgy with industrial procurement and quality assurance.

Start-up and project ideas linked with Unit III

1. **Material-selection assistant:** A simple application that recommends alloy families from density, corrosion, temperature, strength and cost requirements.
2. **Alloy designation learning tool:** A searchable student database for aluminium series, stainless grades and common tool-steel families.
3. **Heat-treatment and ageing calculator:** A teaching tool showing how ageing time/temperature affects precipitation-hardened aluminium alloys.
4. **Cast-iron microstructure classifier:** A basic image-based project to distinguish graphite flakes, nodules and white-iron carbide structures.
5. **Materials certificate checker:** A form-based tool that records grade, heat number, composition, properties and applicable standards for traceability exercises.

High-technology facts related to Unit III

- High-strength steels achieve performance through controlled chemistry, fine grains and small precipitates rather than by carbon alone.
- Modern turbine blades combine nickel-based superalloys, controlled crystal structure, internal cooling and protective coatings.
- Shape memory alloys can act as both sensing and actuating elements through reversible phase transformation.
- Data-driven materials engineering increasingly combines experimental data, simulations and machine learning to screen alloy compositions and processing routes.

Text Books

1. Kenneth G. Budinski and Michael K. Budinski, *Engineering Materials*, Prentice Hall of India Private Limited, 9th Edition, 2018.
2. Sydney H. Avner, *Introduction to Physical Metallurgy*, McGraw Hill Book Company, 1994.

References

1. A. Alavudeen, N. Venkateshwaran, and J. T. Winowlin Jappes, *A Textbook of Engineering Materials and Metallurgy*, Laxmi Publications, 2006.
2. Amandeep Singh Wadhwa and Harvinder Singh Dhaliwal, *A Textbook of Engineering Material and Metallurgy*, University Sciences Press, 2008.
3. G. S. Upadhyay and Anish Upadhyay, *Materials Science and Engineering*, Viva Books Pvt. Ltd., New Delhi, 2020.
4. V. Raghavan, *Materials Science and Engineering*, Prentice Hall of India Pvt. Ltd., 6th Edition, 2019.
5. William D. Callister, *Materials Science and Engineering*, Wiley India Pvt. Ltd., 2nd Edition Reprint, 2019.
6. User-provided course reference: *ME3392 Engineering Materials and Metallurgy – Unit III: Ferrous and Nonferrous Metals*.

Source alignment note: The supplied Unit III handout is used particularly for alloying additions, tool steels, maraging steels, cast irons and copper alloys. Syllabus topics that are not treated in comparable detail in that handout are completed at textbook level using the prescribed references above.

1 Introduction to Unit III

Ferrous and non-ferrous alloys are selected because no single metal can provide every required engineering property. Ferrous alloys are based mainly on iron, while non-ferrous alloys are based on metals such as copper, aluminium, titanium, magnesium and nickel.

The central idea of this unit is:

Composition → Microstructure → Properties → Application

Alloying elements may change phase stability, carbide formation, grain size, hardenability, corrosion resistance, density, thermal stability and wear behaviour. The engineer must therefore connect the role of an element with the final component requirement.

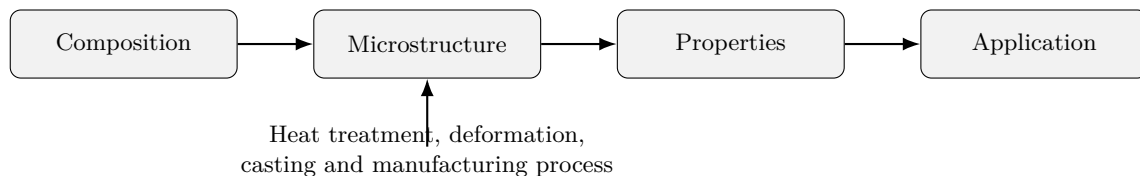


Figure 1: Basic materials-engineering relationship followed throughout Unit III.

Exam point: In most long answers, connect the alloying element or processing route to microstructure, then to properties and finally to applications.

2 Effect of Alloying Additions on Steel

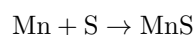
2.1 Why Alloy Plain-Carbon Steel?

Plain-carbon steel is economical and useful, but its properties may be insufficient for severe service. Alloying elements are introduced to obtain one or more of the following:

- increased hardenability and depth of hardening,
- higher strength and wear resistance,
- improved toughness and low-temperature performance,
- corrosion and oxidation resistance,
- high-temperature strength and creep resistance,
- grain refinement and precipitation strengthening,
- improved spring behaviour or hot hardness.

2.2 Manganese (Mn)

Manganese acts as a deoxidiser and combines with sulphur to reduce hot shortness. It also increases strength, hardness and hardenability.



Main effects:

- reduces harmful effect of sulphur during hot working,
- increases tensile strength and wear resistance,
- improves hardenability and response to heat treatment.

Applications: railway components, gears, shafts, wear-resistant steels and many constructional alloy steels.

Remember: Mn → hardenability + sulphur control.

2.3 Silicon (Si)

Silicon is one of the principal deoxidising elements used in steelmaking. It strengthens ferrite and can improve elastic limit and resistance to scaling at elevated temperature.

- deoxidises molten steel,

- increases stiffness and spring properties,
- improves oxidation/scaling resistance in suitable steels,
- excessive silicon can reduce workability and machinability.

Applications: spring steels, electrical steels and selected heat-resistant steels.

Remember: Si → deoxidation + elastic behaviour.

2.4 Chromium (Cr)

Chromium strongly improves hardenability and forms hard carbides. At sufficient level it also enables passive corrosion resistance in stainless steels.

- improves hardenability and heat-treatment response,
- forms hard chromium carbides and increases wear resistance,
- improves oxidation and corrosion resistance,
- improves high-temperature strength in suitable alloy systems.

Applications: stainless steels, bearings, gears, dies, cutting tools and heat-resistant components.

Remember: Cr → hardenability + wear + corrosion resistance.

2.5 Molybdenum (Mo)

Molybdenum is commonly used with other alloying elements. It reduces the critical cooling rate and improves hardenability, elevated-temperature strength and creep resistance.

- increases hardenability,
- improves high-temperature strength and creep resistance,
- reduces tendency to temper embrittlement,
- forms hard carbides and supports tool-steel performance.

Applications: Cr–Mo pressure parts, boiler tubing, shafts, gears and high-speed/tool steels.

Remember: Mo → high-temperature strength + hardenability.

2.6 Nickel (Ni)

Nickel improves toughness, particularly at low temperature, while maintaining useful ductility. It also supports corrosion resistance in high-chromium stainless steels.

- increases toughness and impact resistance,
- improves low-temperature behaviour,
- can reduce quench distortion and broaden heat-treatment range,
- stabilises austenite in stainless steels.

Applications: austenitic stainless steels, cryogenic service, case-hardening steels and high-toughness alloy steels.

Remember: Ni → toughness + low-temperature performance.

2.7 Vanadium (V)

Vanadium is a strong carbide/nitride former and an important grain-refining element.

- forms fine carbides and nitrides,
- refines grain size,
- improves wear resistance and edge retention,
- supports high-temperature strength and hot hardness in tool steels.

Applications: HSLA steels, tool steels, springs, crankshafts and wear-resistant components.

Remember: V → grain refinement + carbide strengthening.

2.8 Titanium (Ti)

Titanium has strong affinity for carbon and nitrogen and forms stable TiC and TiN particles.

- controls carbon and nitrogen,
- restricts grain growth,
- strengthens microalloyed steels,
- stabilises carbon in selected stainless steels and helps avoid chromium-carbide related sensitisation.

Applications: HSLA steels, stabilised stainless grades and fine-grained structural steels.

Remember: Ti $\rightarrow$ C/N control + grain refinement.

2.9 Tungsten (W)

Tungsten is a strong carbide former. Its hard carbides allow steel to retain hardness and cutting ability at elevated temperature.

- increases wear resistance,
- increases hot hardness or red hardness,
- improves cutting power and edge retention,
- inhibits grain growth in suitable tool-steel systems.

Applications: high-speed steels, hot-work steels, drills, milling cutters and other cutting tools.

Remember: W $\rightarrow$ hot hardness + cutting-tool performance.

2.10 Comparison of Alloying Elements

Element	Main effect in steel	Typical engineering link
Mn	Hardenability, strength, sulphur control	Rails, gears, shafts
Si	Deoxidation, ferrite strengthening, spring behaviour	Springs, electrical/heat-resistant steels
Cr	Hardenability, carbides, corrosion/oxidation resistance	Stainless steel, bearings, tools
Mo	High-temperature strength, creep resistance, hardenability	Boilers, pressure parts, tools
Ni	Toughness, low-temperature behaviour, austenite stability	Cryogenic and stainless steels
V	Grain refinement, fine carbides/nitrides	HSLA and tool steels
Ti	TiC/TiN formation, grain control, stabilisation	HSLA, stabilised stainless steel
W	Hot hardness, wear resistance	High-speed cutting tools

Table 2: Summary of the specified alloying additions in steel.

3 Stainless Steels

3.1 Basic Principle

Stainless steels are iron-based alloys containing sufficient chromium to develop a thin, adherent and self-repairing chromium-rich oxide film. This passive film provides their characteristic corrosion resistance.

Stainless behaviour depends on chromium-based passivation; it does not mean that the steel can never corrode.

Nickel, molybdenum, manganese, nitrogen and other elements may be added to control structure and corrosion/mechanical performance.

3.2 Ferritic Stainless Steels

Ferritic stainless steels have a body-centred cubic ferritic matrix over the normal service range.

- mainly chromium-containing steels with low carbon,

- magnetic,
- good resistance to oxidation and stress-corrosion cracking in many environments,
- moderate strength and formability,
- not hardened by conventional quench-and-temper treatment.

Applications: automotive exhaust systems, appliances, architectural panels and selected chemical equipment.

3.3 Martensitic Stainless Steels

Martensitic stainless steels contain enough carbon and suitable chromium to develop martensite after heat treatment.

- heat-treatable to high hardness,
- magnetic,
- good wear resistance,
- corrosion resistance lower than common austenitic stainless steels.

Applications: cutlery, turbine components, surgical instruments, valve parts and wear-resistant shafts.

3.4 Austenitic Stainless Steels

Austenitic stainless steels use nickel and/or other austenite stabilisers to retain FCC austenite at room temperature.

- excellent toughness and formability,
- very good general corrosion resistance,
- generally non-magnetic in the solution-annealed condition,
- cannot be hardened by conventional martensitic quenching, but can strain harden,
- widely used grades include the 18Cr–8Ni family.

Applications: food equipment, chemical vessels, piping, kitchenware, cryogenic equipment and architectural products.

3.5 Duplex and Precipitation-Hardening Stainless Steels – Awareness

Duplex stainless steels contain both ferrite and austenite and combine good strength with resistance to chloride stress-corrosion cracking. Precipitation-hardening (PH) stainless steels obtain high strength by forming fine precipitates during ageing.

Applications: Duplex: chemical/marine process systems. PH stainless: aerospace, shafts, valves and high-strength corrosion-resistant components.

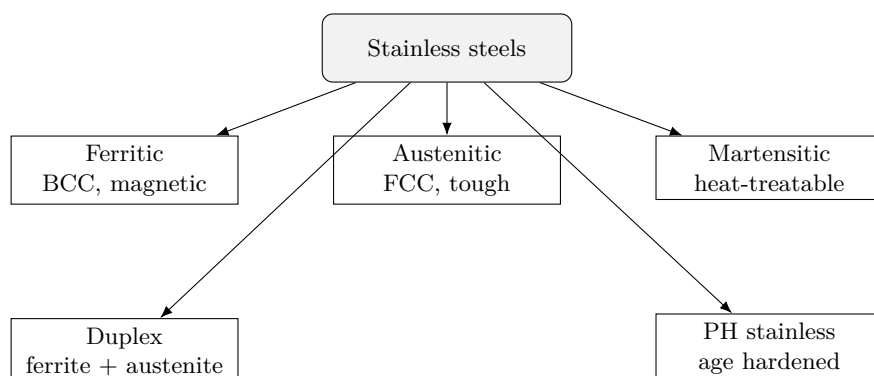


Figure 2: Simple classification of stainless steels.

Exam point: For stainless-steel answers, explain the role of chromium passivation first, then distinguish the matrix/heat-treatment response of each family.

4 Tool Steels

4.1 Definition and Requirements

Tool steels are steels designed for cutting, forming, shearing, moulding or otherwise shaping other materials. They require a suitable combination of hardness, wear resistance, toughness and dimensional stability.

Important requirements include:

- high hardness and wear resistance,
- sufficient toughness to avoid chipping,
- hardenability for the required section size,
- hot hardness for high-speed or hot-work tools,
- resistance to thermal fatigue for dies and moulds.

Carbide-forming elements such as Cr, W, V and Mo are commonly used because stable hard carbides improve wear and edge retention.

4.2 Common Tool-Steel Families

Designation	Main idea	Typical applications
W	Water-hardening steels	Chisels, punches, simple cutters, small tools
O, A, D	Cold-work steels (oil/air/high-alloy families)	Blanking dies, forming dies, shear blades, punches
S	Shock-resistant tool steels	Impact tools, chisels, hammers, shear tools
H	Hot-work tool steels	Hot forging dies, extrusion dies, die-casting dies
T, M	High-speed steels (tungsten or molybdenum types)	Drills, milling cutters, taps, reamers, broaches
P	Plastic-mould steels	Injection moulds and mould bases

Table 3: Important tool-steel families and applications.

4.3 High-Speed Steel and Hot Hardness

During machining, the tool edge becomes hot. High-speed steel retains useful hardness at elevated temperature because of alloy carbides and a heat-treated matrix.

Hot hardness = ability to retain hardness at elevated cutting temperature

Applications: drills, milling cutters, taps, reamers and gear-cutting tools.

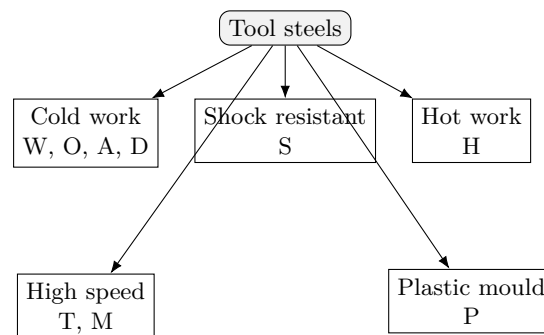


Figure 3: Simplified tool-steel classification for revision.

5 High-Strength Low-Alloy (HSLA) Steels

5.1 Concept

HSLA steels are low-carbon steels designed to obtain higher strength and better service performance than conventional carbon steels through small alloy additions and controlled processing.

Typical microalloying additions include small quantities of Nb, V and Ti. Mn may also be used. Their role is not simply to increase total alloy content; the aim is efficient strengthening with good weldability and formability.

5.2 Strengthening Mechanisms

HSLA performance is obtained through a combination of:

1. **Grain refinement:** smaller ferrite grains increase yield strength and usually improve toughness.
2. **Precipitation strengthening:** fine carbides/nitrides of Nb, V or Ti restrict dislocation motion.
3. **Solid-solution strengthening:** alloy atoms strengthen the ferrite matrix.
4. **Controlled rolling/thermo-mechanical processing:** deformation and cooling are controlled to produce a fine microstructure.

A useful qualitative Hall–Petch relation is:

$$\sigma_y = \sigma_0 + kd^{-1/2}$$

where d is average grain diameter. Smaller grains increase yield strength.

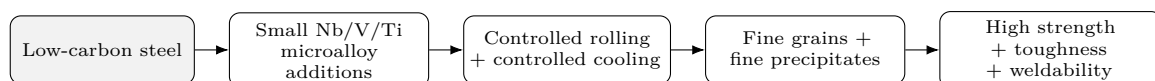


Figure 4: Concept of HSLA steel development.

5.3 Properties and Applications

- high yield strength for relatively low carbon content,
- good weldability compared with high-carbon steels,
- good toughness and fatigue performance,
- weight saving by using thinner sections where design permits.

Applications: automotive structural parts, pipelines, bridges, cranes, construction equipment, pressure structures and railway applications.

Exam point: HSLA does not mean “high alloy”. It is a low-alloy steel strengthened efficiently by grain refinement, precipitation and controlled processing.

6 Maraging Steels

6.1 Basic Principle

Maraging steels are ultra-high-strength, very-low-carbon Fe–Ni steels that derive strength mainly from martensitic transformation followed by precipitation of intermetallic compounds during ageing. Their name combines **martensite + ageing**.

Typical alloying additions include Ni together with Co, Mo and Ti in many grades. Because carbon is very low, the as-quenched martensite is relatively tough and machinable compared with high-carbon martensite.

6.2 Heat-Treatment Cycle

Basic sequence:

Solution treatment → Air cooling → Low-C martensite → Ageing → Intermetallic precipitates

A typical ageing temperature is around 480–500°C for common grades. Fine intermetallic precipitates form during ageing and strongly increase strength.

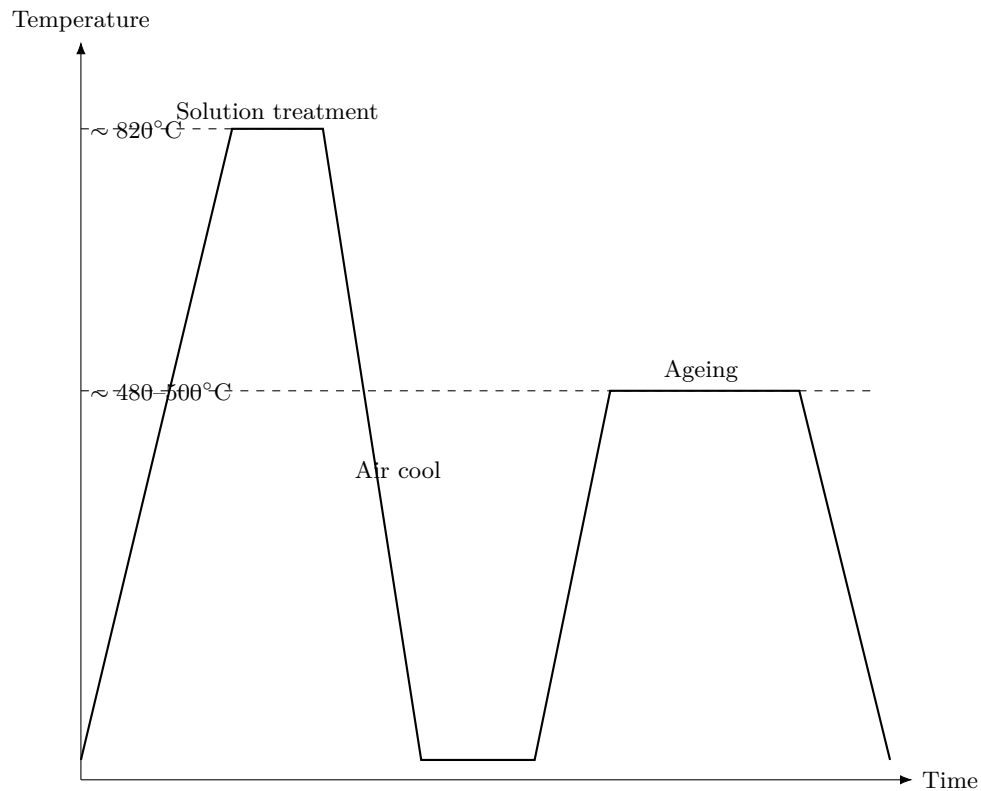


Figure 5: Simplified maraging-steel heat-treatment cycle.

6.3 Properties

- very high strength with good fracture toughness,
- good dimensional stability during ageing,
- good machinability before ageing,
- good weldability compared with many high-carbon ultra-high-strength steels,
- high cost because of alloy content and processing.

Applications: aerospace structures, rocket/missile casings, high-performance shafts and gears, tooling, dies and specialised sporting components.

Point	Conventional high-carbon high-strength steel	Maraging steel
Main strengthening	Carbon martensite/carbides	Low-C martensite + intermetallic precipitation
Machinability before final strengthening	Often difficult	Relatively good
Ageing role	Not primary mechanism	Essential strengthening stage
Typical attraction	Economy and broad availability	Ultra-high strength + toughness + dimensional stability

Table 4: Conceptual comparison of maraging and conventional high-strength steels.

7 Cast Irons

7.1 Why Graphite Form Matters

Cast irons contain higher carbon and silicon than steels and are especially suited to casting. Their mechanical behaviour depends strongly on whether carbon appears as graphite or remains combined as carbide (cementite).

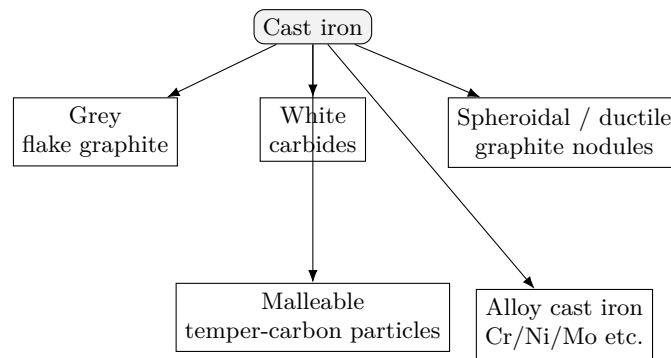


Figure 6: Major cast-iron families in this syllabus.

7.2 Grey Cast Iron

Grey cast iron contains graphite mainly as flakes in a ferritic, pearlitic or mixed matrix. The fracture surface appears grey because of graphite.

- excellent castability and machinability,
- very high vibration damping,
- good compressive strength and self-lubricating tendency,
- graphite flakes act as stress raisers, so tensile ductility and impact strength are low.

Applications: machine beds, engine blocks, brake drums/discs, pump housings and large cast housings.

7.3 White Cast Iron

White cast iron contains carbon mainly in combined carbide form rather than free graphite. It is therefore very hard, abrasion resistant and brittle.

- high hardness and wear resistance,
- poor machinability,
- low impact resistance,
- useful where abrasion is more important than toughness.

Applications: crusher liners, grinding components, rolls and wear plates.

7.4 Malleable Cast Iron

Malleable iron is produced by prolonged heat treatment of white cast iron. Cementite decomposes and carbon appears as compact temper-carbon particles in a ferritic or pearlitic matrix.

- better ductility and shock resistance than grey/white iron,
- good machinability,
- useful for small and medium cast components requiring some toughness.

Applications: pipe fittings, brackets, hubs, small gears, yokes and selected automotive/machine parts.

7.5 Spheroidal Graphite (Ductile/Nodular) Cast Iron

In ductile iron, magnesium or related treatment causes graphite to form approximately spherical nodules rather than flakes.

- nodules create less severe stress concentration than flakes,
- high strength with useful ductility and toughness,
- good machinability and castability,
- properties can be varied by controlling ferritic, pearlitic or heat-treated matrix.

Applications: crankshafts, wheel hubs, gears, valves, pipes, suspension/axle components and agricultural machinery.

7.6 Alloy Cast Irons

Alloy additions such as Cr, Ni and Mo are used when ordinary cast iron cannot provide sufficient wear, heat or corrosion resistance.

- Cr promotes hard carbides and wear resistance,
- Ni can improve matrix toughness and corrosion behaviour,
- Mo can improve hardenability and elevated-temperature performance.

Applications: high-duty rolls, heat-resistant castings, wear parts, pumps and specialised engine components.

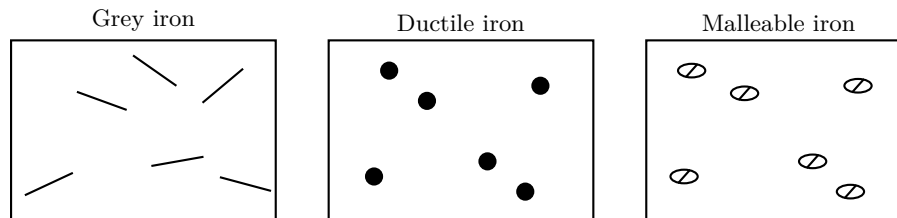


Figure 7: Simplified graphite morphology: flakes, nodules and temper-carbon particles.

Type	Carbon form	Main property	Typical application
Grey	Flake graphite	Damping, machinability, castability	Machine beds, blocks, housings
White	Carbides/cementite	Very hard, abrasion resistant	Crushers, rolls, wear parts
Malleable	Temper carbon	Ductility and shock resistance	Fittings, yokes, small gears
Spheroidal	Graphite nodules	Strength + ductility	Crankshafts, hubs, pipes
Alloy cast iron	Controlled graphite/carbides + alloyed matrix	Heat/wear/corrosion resistance	High-duty specialised castings

Table 5: Comparison of major cast irons.

8 Copper and Its Alloys

8.1 Copper – Engineering Characteristics

Copper is valued for excellent electrical and thermal conductivity, high ductility, corrosion resistance and good joining behaviour. Pure copper has relatively low strength, so alloying is used when improved strength, wear resistance or machinability is needed.

A common trade-off is:

$$\text{Alloying} \uparrow \Rightarrow \text{strength/hardness} \uparrow \quad \text{but electrical conductivity often} \downarrow$$

8.2 Brass – Cu–Zn Alloys

Brass is primarily copper alloyed with zinc.

- zinc increases strength and hardness over pure copper,
- lower-zinc α brasses have excellent cold formability,
- $\alpha + \beta$ brasses are stronger and are commonly hot worked,
- lead may be added to selected brasses to improve machinability.

Applications: plumbing fittings, valves, cartridge cases, fasteners, heat-exchanger parts, decorative products and musical instruments.

8.3 Bronze

Traditionally, bronze means Cu–Sn, although the name also covers aluminium bronze, silicon bronze, phosphor bronze and other copper-based families.

Tin bronze: good wear resistance and bearing properties.

Phosphor bronze: good spring/fatigue behaviour and wear resistance.

Aluminium bronze: high strength and very good marine corrosion resistance.

Silicon bronze: good corrosion resistance and weldability.

Applications: bearings, bushes, gears, springs, marine fittings, pumps, valves and propellers depending on bronze type.

8.4 Cupronickel – Cu–Ni

Cupronickel alloys contain copper with nickel, commonly around 90Cu–10Ni or 70Cu–30Ni families for marine service.

- excellent resistance to seawater corrosion,
- good resistance to erosion-corrosion in flowing seawater,
- useful thermal and fabrication properties.

Applications: marine heat-exchanger/condenser tubes, seawater piping, desalination equipment and coins.

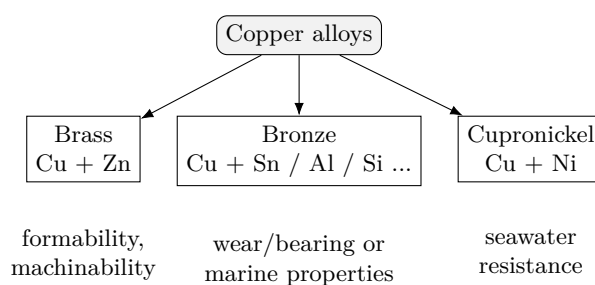


Figure 8: Simple classification and property emphasis of copper alloys.

Alloy	Main addition	Main characteristic	Typical use
Brass	Zn	Formability, machinability, appearance	Valves, fittings, instruments
Tin/phosphor bronze	Sn, P	Wear and spring/bearing behaviour	Bearings, bushes, springs
Aluminium bronze	Al	Strength and marine resistance	Propellers, pumps, valves
Cupronickel	Ni	Seawater corrosion resistance	Marine piping, condensers

Table 6: Comparison of selected copper alloys.

9 Aluminium and Its Alloys

9.1 Why Aluminium?

Aluminium has low density (about 2.7 g/cm^3), good corrosion resistance due to its oxide film, high thermal/electrical conductivity, good formability and excellent recyclability. Pure aluminium is relatively soft, so alloying and strengthening treatments are important.

9.2 Wrought Aluminium-Alloy Series

The first digit of the wrought-aluminium designation identifies the principal alloying family.

Series	Main alloying element	General characteristic / use
1xxx	Commercially pure Al	Conductivity, corrosion resistance, foil and electrical uses
2xxx	Cu	High strength; heat-treatable; aerospace applications
3xxx	Mn	Formability, corrosion resistance; cans and heat exchangers
4xxx	Si	Lower melting range, fluidity; welding/brazing and selected cast-related uses
5xxx	Mg	Marine resistance, weldability; non-heat-treatable structural sheet/plate
6xxx	Mg + Si	Heat-treatable, extrudable; structural and architectural sections
7xxx	Zn (usually with Mg, Cu)	Very high strength; aerospace/high-load applications
8xxx	Other additions	Special-purpose alloys

Table 7: Wrought aluminium-alloy series.

9.3 Effect of Important Alloying Elements in Aluminium

- **Cu:** increases strength and enables precipitation hardening; may reduce corrosion resistance.
- **Mg:** strengthens aluminium, improves work-hardening response and supports marine performance in 5xxx alloys.
- **Si:** improves fluidity and combines with Mg to form the strengthening Mg_2Si system in 6xxx alloys.
- **Mn:** provides moderate strengthening, grain/microstructure control and good formability.
- **Zn + Mg:** produces very high strength in 7xxx alloys through precipitation strengthening.

9.4 Heat-Treatable and Non-Heat-Treatable Families

- **Heat-treatable:** mainly 2xxx, 6xxx and 7xxx series; strengthened by solution treatment, quenching and ageing.
- **Non-heat-treatable:** mainly 1xxx, 3xxx and 5xxx series; strengthened mainly by solid solution and strain hardening.

9.5 Al–Cu Precipitation Strengthening

The purpose is to generate a fine distribution of precipitates that obstruct dislocation motion.

Step 1 – Solution treatment: heat the alloy into a region where Cu-rich phase dissolves in the Al-rich solid solution.

Step 2 – Quenching: cool rapidly to retain a supersaturated solid solution.

Step 3 – Ageing: allow controlled clustering/precipitation at room temperature (natural ageing) or elevated temperature (artificial ageing).

Solution treatment → Quench → Age → Fine precipitates → Higher strength

The strengthening sequence in Al–Cu can be described in simplified form as:



The finest coherent/semi-coherent precipitates provide strong resistance to dislocation motion. With excessive ageing, precipitates coarsen and strength decreases; this is **over-ageing**.

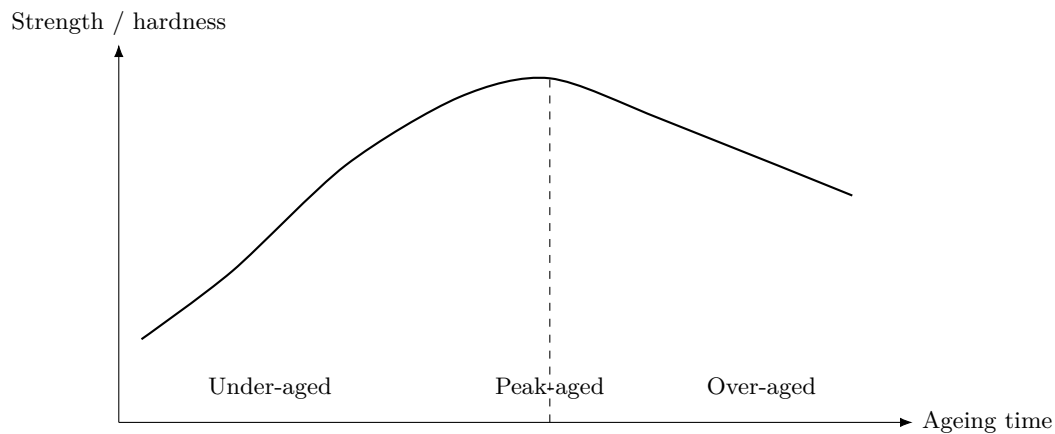


Figure 9: Qualitative effect of ageing time on precipitation-strengthened aluminium.

Applications: 2xxx aircraft structures and fasteners; 6xxx extrusions and structural products; 7xxx high-strength aerospace components.

Exam point: Precipitation hardening is controlled particle formation, not simply “heating to make the metal hard”. Explain solution treatment, quenching and ageing in sequence.

10 Titanium Alloys

10.1 Engineering Importance

Titanium has density around 4.5 g/cm^3 , between aluminium and steel. Its main attraction is high specific strength, excellent corrosion resistance and good fatigue performance.

$$\text{Specific strength} = \frac{\text{strength}}{\text{density}}$$

Titanium forms a stable TiO_2 surface film, giving excellent corrosion resistance in many marine and chemical environments.

10.2 Alpha, Beta and Alpha–Beta Alloys

Pure titanium changes crystal structure with temperature:

- **α titanium:** HCP, stable at lower temperature.
- **β titanium:** BCC, stable at higher temperature.

Alloying elements can stabilise either phase, creating α , β or $\alpha + \beta$ alloy families.

Alpha stabilisers: Al is the most important common example.

Beta stabilisers: V, Mo, Nb and others can stabilise the beta phase.

10.3 Ti–6Al–4V

Ti–6Al–4V is a widely used $\alpha + \beta$ titanium alloy containing approximately 6% Al and 4% V.

- high strength-to-weight ratio,
- good fatigue resistance,
- good corrosion resistance,
- useful balance of processability and performance.

Applications: aircraft structures, compressor components, high-performance fasteners, chemical equipment and biomedical devices.

10.4 Limitations

- high raw-material and processing cost,
- difficult machining and high tool wear,

- reactive at high processing temperatures,
- lower elastic modulus than steel, so stiffness must be considered.

Remember: Titanium is selected when high performance justifies higher cost.

11 Magnesium Alloys

11.1 Key Feature – Very Low Density

Magnesium has density about 1.74 g/cm^3 and is the lightest commonly used structural metal.

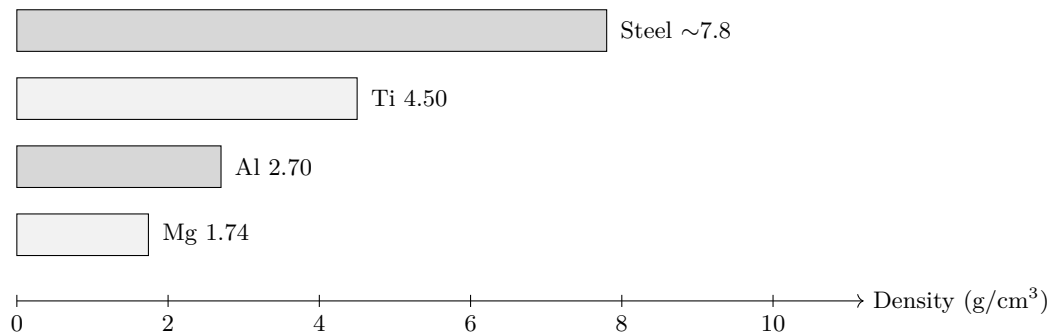


Figure 10: Approximate density comparison of common structural metals.

11.2 Properties

- very low density and good specific strength,
- excellent machinability and die-castability,
- good vibration damping,
- useful electromagnetic shielding for electronic housings,
- corrosion and high-temperature performance require careful alloy/surface design.

Common alloying additions include Al, Zn, Mn and selected rare-earth elements. Al improves strength and castability; Zn supports strengthening; Mn helps corrosion control by reducing the effect of harmful impurities.

11.3 Designation Example – AZ91

In a common magnesium designation, letters identify major alloying elements and numbers indicate their approximate percentages. For **AZ91**, A represents aluminium and Z represents zinc; it contains roughly 9% Al and 1% Zn.

Applications: gearbox and transmission housings, steering components, seat frames, laptop/camera bodies, portable tools and selected aerospace housings.

11.4 Safety Note

Bulk magnesium components are not normally easy to ignite in service. However, fine chips, dust and powder have high surface area and can burn rapidly. Magnesium machining therefore requires proper chip control, housekeeping and appropriate Class-D fire precautions.

12 Nickel-Based Superalloys

12.1 Why Superalloys?

Gas turbines, aero-engines and other hot-section systems require materials that retain strength while resisting creep, oxidation and thermal cycling.

High temperature + high stress + long exposure → need for superalloys

Nickel is a useful base because it has an FCC matrix, broad alloying capability, good phase stability and useful high-temperature strength.

12.2 Basic Microstructure

A simplified nickel-based superalloy microstructure contains:

$$\gamma \text{ matrix} + \gamma' \text{ strengthening precipitates}$$

- **γ matrix:** Ni-rich FCC solid solution; provides ductility and toughness.
- **γ' :** ordered precipitates based approximately on $\text{Ni}_3(\text{Al,Ti})$; strongly restrict dislocation motion and provide high-temperature strength.

Typical alloy roles:

- Cr – oxidation/corrosion resistance,
- Al/Ti/Ta – precipitation strengthening and protective oxides,
- Mo/W – solid-solution strengthening,
- Co – high-temperature stability,
- B/C/Zr – grain-boundary effects in suitable grades.

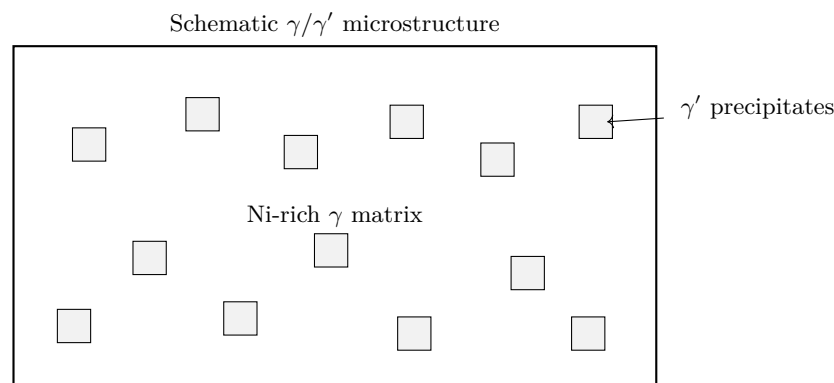


Figure 11: Simplified view of fine γ' precipitates in a γ matrix.

12.3 Creep and Turbine-Blade Design

Creep is slow time-dependent deformation under sustained stress, especially at high temperature.

$$\text{Creep behaviour} = f(\text{stress, temperature, time})$$

Advanced turbine blades may use directionally solidified or single-crystal structures to reduce harmful grain-boundary effects at high temperature. Internal cooling passages and thermal-barrier coatings are also used.

Applications: aero-engine turbine blades/discs, gas-turbine hot-section components, rocket systems, high-temperature fasteners and chemical/process equipment.

13 Shape Memory Alloys

13.1 Basic Principle

Shape memory alloys (SMAs) can recover large deformation through a reversible solid-state phase transformation. The best-known engineering SMA is Ni-Ti (Nitinol).

Two important behaviours are:

1. **Shape memory effect** – heating restores a previously trained shape.
2. **Superelasticity** – large strain is recovered on unloading through stress-induced transformation.

13.2 Martensite–Austenite Transformation

$$\text{Martensite} \rightleftharpoons \text{Austenite}$$

Martensite is stable at lower temperature and can be reoriented/deformed relatively easily. Austenite is stable at higher temperature and corresponds to the trained shape.

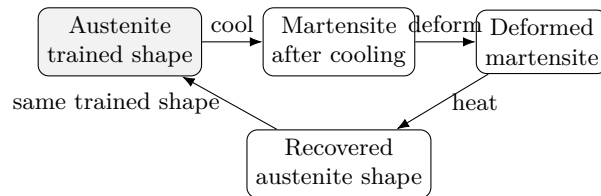


Figure 12: Simplified shape-memory cycle.

13.3 Shape Memory Effect and Superelasticity

Feature	Shape memory effect	Superelasticity
Main trigger	Temperature change	Mechanical stress/unloading
Initial phase	Martensitic condition	Austenitic condition above transformation range
Recovery	Heating restores trained shape	Unloading restores shape
Example	Thermal actuator/coupling	Eyeglass frame or vascular stent

Table 8: Difference between shape memory effect and superelasticity.

13.4 Applications

- orthodontic wires – nearly continuous corrective force,
- vascular stents – superelastic expansion and shape recovery,
- eyeglass frames – large recoverable deformation,
- temperature-controlled valves and switches – thermally activated motion,
- pipe couplings – recovery force provides clamping,
- aerospace/robotic actuators – compact, silent actuation,
- vibration-control devices – energy absorption through transformation.

Limitations include high material cost, non-linear behaviour, sensitivity to heat treatment/composition, slower thermal response and functional fatigue under repeated cycling.

14 Overview of Materials Standards

14.1 Why Standards Are Needed

A material description such as “high-strength steel” or “good aluminium” is not adequate for engineering procurement. Standards create a common technical language between designer, manufacturer, supplier, testing laboratory and customer.

Materials standards may specify:

- chemical composition,
- mechanical properties,
- product form and dimensions,
- heat-treatment/temper condition,
- test method and specimen requirements,
- inspection and acceptance criteria,
- marking, certification and traceability.

14.2 Common Standards Organisations

Organisation	Role / context
ISO	International standards across many engineering fields
ASTM	Material specifications and test methods widely used internationally
BIS	Bureau of Indian Standards; Indian national standards
SAE	Automotive/aerospace material and engineering specifications
ASME	Codes and standards for boilers, pressure vessels and mechanical systems
EN	European Standards
JIS	Japanese Industrial Standards
DIN	German standards system / national standards body context
AWS	Welding standards and procedures
API	Petroleum and process-industry standards

Table 9: Examples of standards organisations encountered by mechanical engineers.

14.3 Material Grade, Specification and Test Method

Material grade/designation identifies the alloy, for example 6061 aluminium, AISI 304 stainless steel or Ti-6Al-4V.

Material specification defines acceptable composition, properties, product condition and other requirements.

Test-method standard defines how a property is measured so that results are comparable.

Grade tells us what material; the standard/specification tells us the required technical conditions.

14.4 Material Certificate and Traceability

A material certificate may include grade, heat/batch number, chemical composition, mechanical properties, heat-treatment condition, test results, applicable standard and manufacturer identification.

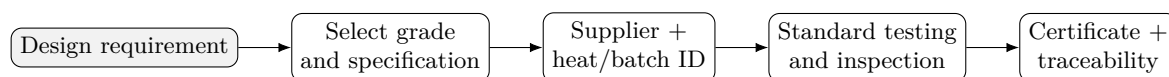


Figure 13: How standards connect material selection, testing and traceability.

Exam point: A measured strength value is meaningful only when the material condition and test method are known.

15 Important Exam-Oriented Points

15.1 Very Short Answer Points

1. Mn improves hardenability and combines with sulphur to reduce hot shortness.
2. Si is an important steel deoxidiser and improves spring/elastic behaviour.
3. Cr improves hardenability, wear resistance and corrosion/oxidation resistance.
4. Mo improves hardenability, high-temperature strength and creep resistance.
5. Ni improves toughness and low-temperature performance.
6. V and Ti are important grain-refining and carbide/nitride-forming additions.
7. W provides hot hardness and wear resistance in high-speed steels.
8. Stainless steels obtain corrosion resistance mainly through chromium-based passivation.
9. Tool steels are designed for hardness, wear resistance and sufficient toughness.
10. HSLA steels use low carbon, microalloying and controlled processing for high strength.
11. Maraging steel = low-carbon martensite strengthened by ageing of intermetallic precipitates.
12. Grey iron contains graphite flakes and has excellent damping/machinability.

13. White iron contains mainly carbide and is very hard and brittle.
14. Malleable iron is heat-treated white iron containing temper-carbon particles.
15. Spheroidal graphite iron contains graphite nodules and has high strength with useful ductility.
16. Brass is mainly Cu–Zn; bronze is mainly Cu–Sn or related Cu-base families; cupronickel is Cu–Ni.
17. The 2xxx Al series is Cu-based; 5xxx is Mg-based; 6xxx is Mg–Si; 7xxx is Zn-based.
18. Precipitation hardening consists of solution treatment, quenching and ageing.
19. Ti–6Al–4V is a common $\alpha + \beta$ titanium alloy.
20. Magnesium is the lightest commonly used structural metal.
21. Nickel-based superalloys are used for high-temperature strength and creep resistance.
22. Nitinol is a Ni–Ti shape memory alloy.
23. Shape memory effect is thermally activated; superelastic recovery occurs on unloading.
24. Materials standards support consistent composition, testing, inspection and traceability.

15.2 Important Two-Mark Questions

1. State any two purposes of alloying steel.
2. What is the role of Mn in controlling sulphur in steel?
3. State two effects of Si in steel.
4. Why is chromium important in stainless steel?
5. State two effects of Mo in steel.
6. Why is nickel used in low-temperature steels?
7. How do V and Ti refine/strengthen steel?
8. What is meant by hot hardness? Which alloying element is important for it?
9. Define stainless steel and explain passivation briefly.
10. List the principal families of tool steel.
11. Define HSLA steel.
12. What is maraging steel?
13. Why does grey cast iron have good damping capacity?
14. Differentiate grey and white cast iron.
15. What is malleable cast iron?
16. What is spheroidal graphite cast iron?
17. Define brass, bronze and cupronickel.
18. List the main wrought-aluminium series from 2xxx to 7xxx with principal alloying elements.
19. State the three steps in precipitation strengthening of Al–Cu alloy.
20. What is over-ageing?
21. State two important properties of titanium alloys.
22. Why is magnesium important for lightweight engineering?
23. What are γ and γ' in nickel-based superalloys?
24. Define shape memory effect and superelasticity.
25. What is a material standard? Why is traceability important?

15.3 Important Five-Mark Questions

1. Explain the effects of Mn, Si, Cr and Mo in steel with applications.
2. Explain the effects of Ni, V, Ti and W in steel with applications.
3. Compare ferritic, martensitic and austenitic stainless steels.
4. Classify tool steels and state their applications.
5. Explain the concept, strengthening mechanisms and applications of HSLA steels.
6. Explain the heat treatment, properties and applications of maraging steels.
7. Compare grey, white, malleable and spheroidal cast irons.
8. Explain brass, bronze and cupronickel with properties and applications.
9. Explain wrought-aluminium alloy series and the role of major alloying elements.
10. Explain Al–Cu precipitation strengthening with ageing curve.
11. Explain the classification and applications of titanium alloys.
12. Explain properties, alloying and applications of magnesium alloys.
13. Explain the microstructure and applications of nickel-based superalloys.
14. Differentiate shape memory effect and superelasticity with applications.
15. Explain the need for materials standards and material traceability.

15.4 Important Long-Answer Questions

1. Discuss the effect of Mn, Si, Cr, Mo, Ni, V, Ti and W on the properties and applications of steel.
2. Explain stainless steels, tool steels, HSLA steels and maraging steels with suitable properties and applications.
3. Explain the structure, properties and applications of grey, white, malleable, spheroidal and alloy cast irons.
4. Explain copper and its important alloys – brass, bronze and cupronickel – with properties and engineering applications.
5. Explain aluminium alloy classification and Al–Cu precipitation strengthening treatment with a neat ageing curve.
6. Explain titanium alloys, magnesium alloys and nickel-based superalloys with emphasis on property–application relationships.
7. Explain shape memory alloys, shape memory effect and superelasticity with real engineering applications.
8. Give an overview of materials standards, specifications, testing standards and traceability in engineering practice.

16 Unit Summary

- Alloying additions modify steel hardenability, toughness, corrosion resistance, grain size, wear and high-temperature behaviour.
- Mn controls sulphur and hardenability; Si deoxidises and strengthens ferrite; Cr improves hardenability/wear/corrosion; Mo improves elevated-temperature strength; Ni improves toughness; V/Ti refine and precipitate; W gives hot hardness.
- Stainless steels depend on chromium passivation; matrix type gives ferritic, austenitic and martensitic families.
- Tool steels are selected according to cold work, hot work, shock, high-speed and mould requirements.
- HSLA steels combine low carbon, fine grains, precipitation and controlled processing.
- Maraging steels obtain ultra-high strength from low-C martensite followed by ageing of intermetallic precipitates.
- Graphite/carbide morphology controls cast-iron behaviour: flakes in grey iron, carbides in white iron, temper carbon in malleable iron and nodules in ductile iron.
- Brass is Cu–Zn, bronze is a Cu-base family commonly involving Sn/Al/Si, and cupronickel is Cu–Ni.

- Aluminium-alloy series identify principal alloying families; 2xxx, 6xxx and 7xxx are important heat-treatable systems.
- Al–Cu precipitation strengthening uses solution treatment, quenching and ageing; over-ageing reduces strength by precipitate coarsening.
- Titanium provides high specific strength and corrosion resistance; magnesium provides extremely low density and good castability.
- Nickel-based superalloys use a γ matrix strengthened by γ' precipitates for severe high-temperature service.
- Shape memory alloys use reversible martensite–austenite transformation to provide shape recovery and superelasticity.
- Materials standards establish common rules for grade, properties, test methods, inspection and traceability.

17 Glossary of Important Terminologies

The following short definitions are useful for quick revision and two-mark examination answers.

Term	Short definition / exam point
Alloy steel	Steel containing intentional alloying additions to obtain properties beyond plain-carbon steel.
Hardenability	Ability of steel to develop martensitic hardness to depth during quenching.
Hot shortness	Brittleness/cracking tendency during hot working, commonly associated with low-melting sulphide films when sulphur is not properly controlled.
Passivation	Formation of a thin protective oxide film that greatly reduces corrosion rate.
Stainless steel	Chromium-containing steel designed to form a protective passive film and resist corrosion.
Ferritic stainless steel	Stainless steel with mainly ferritic BCC matrix at room temperature.
Martensitic stainless steel	Heat-treatable stainless steel capable of forming martensite and high hardness.
Austenitic stainless steel	Stainless steel with FCC austenitic matrix, usually stabilised by Ni and/or other austenite stabilisers.
Tool steel	Steel designed for cutting, forming, moulding or shaping other materials.
Hot hardness	Ability to retain hardness at elevated temperature.
HSLA steel	High-strength low-alloy steel using low carbon, small alloy additions and controlled processing.
Microalloying	Addition of very small amounts of elements such as Nb, V or Ti for grain/precipitation strengthening.
Maraging steel	Very-low-carbon Fe–Ni steel strengthened by martensite formation followed by ageing of intermetallic precipitates.
Grey cast iron	Cast iron containing graphite mainly as flakes.
White cast iron	Cast iron in which carbon is mainly in combined carbide form, producing high hardness and brittleness.
Malleable cast iron	Heat-treated white cast iron containing compact temper-carbon particles.
Spheroidal graphite iron	Ductile/nodular cast iron containing approximately spherical graphite nodules.
Brass	Copper–zinc alloy family.
Bronze	Copper-base alloy family traditionally based on Cu–Sn and including aluminium/silicon/phosphor bronzes.
Cupronickel	Copper–nickel alloy family with good seawater resistance.
Specific strength	Strength divided by density; useful for lightweight design comparison.
Solution treatment	Heating into a single-phase region to dissolve strengthening solute before quenching.
Supersaturated solid solution	Non-equilibrium solid solution containing more solute than equilibrium allows at that temperature.
Ageing	Holding after quenching so that clusters/precipitates form and change strength.
Natural ageing	Ageing at approximately room temperature.
Artificial ageing	Ageing at elevated temperature below the solution-treatment temperature.
Over-ageing	Strength reduction caused by precipitate coarsening or transformation during excessive ageing.
α titanium	HCP titanium phase stable at lower temperature.

β titanium	BCC titanium phase stable at higher temperature and stabilised by suitable alloying elements.
Superalloy	High-performance alloy designed to retain strength and environmental resistance at elevated temperature.
γ matrix	Ni-rich FCC matrix phase in many nickel-based superalloys.
γ' precipitate	Ordered $\text{Ni}_3(\text{Al,Ti})$ -type strengthening precipitate in many nickel-based superalloys.
Creep	Time-dependent permanent deformation under sustained stress, important at elevated temperature.
Shape memory alloy	Alloy capable of recovering large deformation through reversible phase transformation.
Shape memory effect	Recovery of trained shape on heating through a transformation range.
Superelasticity	Large recoverable strain caused by stress-induced transformation and reverse transformation on unloading.
Material grade	Designation identifying a defined alloy/material family or composition range.
Material specification	Technical document defining required composition, properties, product condition and acceptance requirements.
Test-method standard	Standardised procedure defining how a property is measured.
Heat number	Batch identifier connecting a product to a specific melt/production heat for traceability.
Traceability	Ability to connect a component/material to its origin, batch, processing and inspection records.
